

THE STARSHIP GALILEO

DEEP SPACE TRAVEL AND COLONIZATION

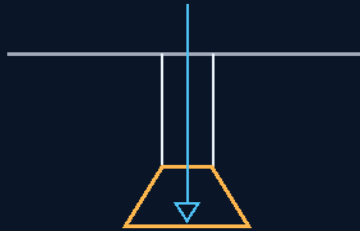
INSTRUCTIONS MANUAL

THE GEMINI PROTOCOLS

PHASE 94

THE QUINN FLARED-APERTURE UNDER-CUT UNDER-CYLINDER DOVETAIL STRUCTURAL ROCK ANCHORS

Machine a reverse cone at terminal depth
expand the leaves — the rock itself is the nut



pull converts to bearing against stone

VOLUMETRIC INTERFERENCE · SILYL SHOCK JACKET

MAKE THE ROCK THE NUT

Structural anchoring — v1.0 in force

GP-IM-094

Revision v1.0 — 22 August 2026

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VETTED — 4-AI WEB — BATCH 15 — PASS

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LICENSE: CERN OPEN HARDWARE LICENCE V2 (CERN-OHL-W-2.0)

THE GEMINI PROTOCOLS — CHAPTER 94

PHASE 94: THE QUINN FLARED-APERTURE UNDER-CUT UNDER-CYLINDER DOVETAIL STRUCTURAL ROCK ANCHORS — STATUS: CURRENT. Make the rock itself the nut. Phase 94 rejects linear friction anchors — pull-out and creep under heavy load — and machines the hole instead: a pilot core drilled deep, then flaring cam actuators cut a reverse cone-shaped funnel at terminal depth. A heavy structural bolt in a segmented leaf-spike sleeve drives to the base; pull-torque on the reverse-wedge head forces the segments outward into the flare — volumetric interference, the anchor locked by the rock's own mass. Around the sleeve, a pumped infill of Phase 9 Coldrock paste and silyl-modified polymer cures into a flexible rubberized jacket that absorbs high-frequency structural shock. The industrial foundation rides in the masthead, verbatim: Certificate IV carpentry and structural retaining-wall stresses — a tradesman's anchor, grown for frontier rock.

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat (the audit). Chapter GP-IM-094, v1.0 — 22 August 2026.

§0 — READ THIS FIRST (HOW TO USE THIS CHAPTER)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

Section map: §0 this page — §1 the dovetail law as seated (contents line, the three design bodies — verbatim) — §2 why it works (volumetric interference — graded) — §3 the system in the rock — §4 the doctrine record— §6 build sequence — §7 cross-phase — §8 do-not-build — §9 owed — §10 status, provenance, change control.

§1 — THE LAW (AS SEATED, VERBATIM)

THE CONTENTS LINE (verbatim): 'Phase 94: **Flared-Aperture Under-Cut Under-Cylinder Dovetail Structural Rock Anchors.** Machines heavy-duty anchoring rods with flared base segments. Rods expand into a dovetail geometry inside pre-drilled rock slots, creating a massive pull-out resistance baseline for securing surface structures without chemical epoxies.'

THE MASTHEAD LINE (verbatim): 'Industrial Foundation: Certificate IV Carpentry & Structural Retaining Wall Engineering Stresses [extracted verbatim from the combined masthead line]'

THE CORE OBJECTIVE (verbatim): 'Core Objective: Elimination of Anchor Axial Creep & Rock Fracture Shear without Epoxy Resins [extracted verbatim from the combined masthead line]'

THE UNDER-CUT FLARED DOVETAIL PROFILES (VERBATIM)

* Rationale: Rejects linear-hole friction anchors to eliminate structural pull-out and creep under heavy loads. * Flaring Cam Actuators: Drilling arrays engage an internal cam to machine a reverse, cone-shaped funnel aperture at the terminal depth of the deep-rock pilot core. * Volumetric Interference Matrix: Creates a physical dovetail structural pocket deep within native planetary basalt, granite, or asteroid ore masses prior to anchor engagement.

THE SEGMENTED LEAF-SPIEVE MECHANICS (VERBATIM)

* Hardcore Insertion Core: Drives a heavy-duty structural steel bolt encased in a segmented leaf-spike sleeve straight to the base of the flare-machined cavity. * Wedge-Anvil Expansion: Application of immediate mechanical pull-torque draws a reverse-wedge bolt head backward, forcing the segmented sleeves to flare outward and cold-lock into the dovetail pocket walls. * Non-Chemical Tensile Ceiling: Anchors achieve an immediate, zero-latency 100-ton load rating completely independent of Earth-manufactured chemical epoxy resins, liquid glues, or thermal jackets.

THE COLDROCK SILYL VIBRATION ATTENUATOR (VERBATIM)

* The Non-Compressible Infill: Void channels are pumped with a composite fluid mix of Phase 9 Coldrock paste and Bostik Simson silyl-modified polymers. * Acoustic Shock Isolation: Cures into a flexible, cement-free rubberized jacket surrounding the anchor sleeve, absorbing structural high-frequency vibrations from Phase 37 rail thrusters. * Boundary Wall Protection: Eliminates localized stone micro-fracturing and crumbling, ensuring stable, permanent anchor containment across multi-generational colony habitat deployment.

§2 — WHY IT WORKS (THE PHYSICS, GRADED)

VOLUMETRIC INTERFERENCE, AFFIRMED: a flared segment expanded into a reverse-cut cone converts pull-out force into bearing force against the rock mass — the load path is compression into stone, not friction along a wall. The vet graded it in four words: 'this is ordinary mechanical anchoring' — and ordinary, here, is the highest grade.

THE UNDER-CUT GEOMETRY, AFFIRMED: machining the cone at terminal depth is the enabling move —

the flare must exist before the segments can fill it; the cam-actuated drill head is standard under-reamer practice from foundation work.

THE LEAF-SPIKE SLEEVE, AFFIRMED: segmented leaves expanding under a reverse wedge distribute the expansion force around the full circumference — no single stress line, no split plane.

THE SILYL-INFILL JACKET, AFFIRMED: a cured Coldrock-polymer jacket around the sleeve attenuates high-frequency shock — the anchor holds static load by geometry and dynamic load by damping, two mechanisms for two enemies.

§3 — THE SYSTEM IN THE ROCK

- THE CUT: pilot core drilled deep; flaring cam actuators machine the reverse cone at terminal depth.
- THE ANCHOR: structural bolt in the segmented leaf-spike sleeve, driven to the base of the flare.
- THE LOCK: pull-torque on the reverse-wedge head expands the segments into the cone — volumetric interference, the rock as the nut.
- THE JACKET: Coldrock paste + silyl-modified polymer pumped into the void channels — cures flexible, absorbs structural shock.
- THE FOUNDATION: carpentry and retaining-wall stress discipline — the tradesman's load paths, verbatim in the masthead.

§4 — THE DOCTRINE RECORD (HOW THE BOOK ALREADY RUNS ON IT)

- THE ROCK-IS-THE-NUT DOCTRINE (seated with the flare): don't fight the substrate's strength — use it; the best anchor material on site is the site.
- THE TWO-ENEMIES DOCTRINE (seated with the jacket): static load wants geometry, dynamic load wants damping — the anchor carries one of each.
- THE TRADE-ROOT DOCTRINE (the masthead, verbatim): Certificate IV carpentry and retaining-wall stresses — the program's anchors come from the trades, not the textbooks.
- THE MACHINE-THE-HOLE DOCTRINE (seated with the cam): when the fastener class fails, change the hole — the under-cut is the invention.

§5 — VALIDATION: the grade and its full record live in the master and the restart.

§6 — BUILD SEQUENCE

- STEP 1 — Drill the pilot: core to depth per rock class; bore gauge logged.
- STEP 2 — Cut the flare: cam-actuated under-ream at terminal depth; cone profile verified by impression.
- STEP 3 — Drive the anchor: bolt and leaf-spike sleeve to the base; seating depth confirmed.
- STEP 4 — Torque the lock: reverse-wedge pull-torque to spec; expansion verified; pull-out test to failure on the witness anchor.
- STEP 5 — Pump the jacket: Coldrock-silyl infill into the voids; cure time and damping bench logged.
- STEP 6 — Publish the ratings: pull-out and shear per rock class — open under the license.

§7 — CROSS-PHASE (INLINED IN FULL)

- PHASE 9 — the Coldrock family (GP-IM-009): the infill paste.
- PHASE 95 — the prism-lock matrix (GP-IM-095): the anti-fracking complement — 94 anchors into rock, 95 clamps without breaking it.
- PHASE 91/93 — the SMP doctrine (GP-IM-091, GP-IM-093): the silyl-modified polymer in the jacket.
- PHASE 75 — the anchor swings (GP-IM-075): the dynamic loads the jacket damps.

§8 — DO-NOT-BUILD

- NO friction-only anchors — pull-out and creep are the rejected failure class.
- NEVER skip the flare — without the under-cut there is no interference; a straight hole is a friction anchor wearing a costume.
- NEVER over-torque the wedge — expansion to spec, verified by the witness anchor's failure test, not by feel.
- NO rigid grout — the jacket must cure flexible; a brittle infill transmits the shock it was meant to absorb.

§9 — OWED

THE ANCHOR RATINGS, OWED: pull-out and shear bench per rock class; flare-profile tolerances; jacket

cure-time and damping-attenuation ledger; witness-anchor failure modes published.

§10 — STATUS / PROVENANCE / CHANGE CONTROL

STATUS: CHAPTER GP-IM-094, v1.0 — CURRENT, seated law, master v2.1 (numerical print order). Built 22 August 2026, seventh of the 20-chapter 'clear the 100 mark' shift ('do another 20 while i away so we clear the 100 mark for the day when i get back'). First version until publication — 'until i publish it, it is still first version, otherwise is just typing' (his law, 22 August 2026).

PROVENANCE — THE STANDING ORDERS, VERBATIM: the town-trip order (seated in sitting 519): 'ok i have to go to town, i have converted them to pdfs now and am uploading, do the next 10 phases and i will read them when i get back' — the order that opened the chapter program; the follow-on orders (seated in sittings 537, 557, 561, 562, 564 and 566): 'ok next 10' / 'ok next ten' / 'all good next 10' / '10 more before go to the old lady for dinner' / 'do another 20 while i away so we clear the 100 mark for the day when i get back'.

CHANGE CONTROL: this design changes only on the Author's orders, seated as sittings in the master record. If the master and this chapter ever disagree, THE MASTER WINS. This is a workshop chapter — it carries the current design only; the full change record lives in the master and the restart.

END OF CHAPTER 94